REAL TIME SAFETY MONITORING SYSTEM FOR PUBLIC TRANSPORT IN SRILANKA

Project Group: 25-26J-499

Project Proposal Report

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Sri Lanka

August 2025

DROWSINESS DETECTION AND SEATBELT USAGE

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor:

Date 25/08/2025

ABSTRACT

Road safety remains a critical concern in Sri Lanka's public transport sector, where driver fatigue and inadequate safety protocol adherence contribute significantly to accident rates. This research proposes developing an innovative, cost-effective safety monitoring system that simultaneously addresses two primary risk factors: driver drowsiness and seatbelt non-compliance. The system uniquely combines computer vision-based drowsiness detection with physical sensor-based seatbelt monitoring on a unified Arduino Uno platform.

The drowsiness detection component employs facial landmark analysis utilizing Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and head pose estimation techniques. These methods collectively identify fatigue indicators including prolonged eye closure, excessive yawning, and abnormal head movements. A complementary seatbelt verification module uses embedded sensors to ensure compliance with safety regulations.

What distinguishes this solution is its complete offline operation capability, eliminating dependency on internet connectivity while maintaining real-time performance. The system provides immediate multi-modal alerts through auditory buzzers, visual LED indicators, and display messages, ensuring drivers receive prompt warnings. By implementing this integrated approach on affordable hardware, the research addresses the critical gap between advanced safety technologies and practical deployment in Sri Lanka's public transport infrastructure.

The anticipated outcome is a robust, economically viable safety system that can significantly reduce accident risks while being accessible for widespread implementation across public transport fleets.

Keywords: Driver safety, drowsiness detection, seatbelt monitoring, public transport, Arduino, computer vision

TABLE OF CONTENT

Contents

DECLARATION 3

ABSTRACT 4

TABLE OF CONTENT 5

LIST OF FIGURES 6

1. INTRODUCTION 7

1.1 BACKGROUND & LITERATURE SURVEY 8

1.2 RESEARCH GAP 12

1.3 RESEARCH PROBLEM 15

2. OBJECTIVES 17

2.1 Research Questions 17

2.2 Main Objectives 18

2.3 Specific Objectives 18

3. METHODOLOGY 19

4. PROJECT REQUIREMENTS 25

4.1 Functional Requirements 23

4.2 Non-Functional Requirements 24

4.3 Expected Test Cases 25

5. COMMERCIALIZATION POTENTIAL 29

6. WORK BREAKDOWN STRUCTURE 32

7. REFERENCES 34

8. APPENDICES 35

LIST OF FIGURES

	page
Figure 2.1: System Architecture of the Driver Safety Monitoring System	19
Figure 2.2: 68 landmarks that are detection	20
Figure 2.3: Detailed Workflow of the Drowsiness Detection Module	22
Figure 2.4: Communication Flow between Arduino, Camera, and Alert System	23
Figure Appendices : Turnitin Score	

LIST OF TABLES

	page
Table 1: Research gap table	12
Table 8.1: Market comparison with existing systems	29
Table 6.1: Project Timeline table	32
Table 6.2: Project budget breakdown	33

LIST OF ABBREVIATIONS

ABBREVIATIONS	DESCRIPTION
AI	Artificial Intelligence
CV	Computer Vision
EAR	Eye Aspect Ratio

1. INTRODUCTION

Public transport in Sri Lanka serves millions daily but contributes significantly to the nation's road accident statistics. Each year, thousands of fatalities and injuries are reported, with driver-related factors such as fatigue, distraction, and non-compliance with safety protocols—like seatbelt usage—identified as primary causes. Existing enforcement methods rely heavily on manual oversight, which is often ineffective and resource intensive. Therefore, a shift toward advanced, proactive technological interventions is necessary.

Research in driver safety has evolved from physiological signal monitoring to AI-driven visual monitoring of drivers. Key techniques include facial landmark detection, Eye Aspect Ratio (EAR) for eye closure detection, Mouth Aspect Ratio (MAR) for yawning detection, and head pose estimation to detect nodding or distraction. Current state-of-the-art solutions combine these metrics with machine learning algorithms to improve accuracy and robustness under varied conditions. Many systems also integrate multiple sensors to provide comprehensive safety assessments.

However, most existing solutions face practical deployment challenges in Sri Lanka's public transport sector, such as high computational requirements, reliance on stable internet connectivity, and cost. Additionally, systems designed for private vehicles often do not account for the unique operational environment of public buses, including vibrations, varying lighting, and the need for localized alerts.

This research proposes an integrated Safety Monitoring System specifically tailored for Sri Lankan public transport. It focuses on four core components: driver monitoring (drowsiness and seatbelt usage), rule violation detection, real-time passenger monitoring with ID-based entry, and live bus tracking.

This proposal details the driver monitoring component—utilizing facial landmark analysis and physical seatbelt sensors implemented on a low-cost Arduino Uno platform—to provide immediate offline alerts. This system aims to reduce accident rates and foster enhanced driver awareness and passenger security.

1.1. BACKGROUND & LITERATURE SURVEY

1.1.1 Background

Sri Lanka's public transportation infrastructure serves millions of passengers daily, yet remains plagued by safety challenges that result in significant human and economic costs. Statistical analysis of accident data reveals that driver fatigue contributes to approximately thirty percent of serious traffic incidents involving public transport vehicles [11]. Simultaneously, non-compliance with basic safety protocols, including seatbelt usage, exacerbates injury severity when accidents occur.

The economic impact of these safety issues extends beyond immediate medical costs and property damage. Public confidence in transportation safety affects ridership patterns, while insurance costs for transport operators continue rising due to frequent claims. Government expenditure on emergency response and healthcare treatment for accident victims represents a substantial drain on public resources that could be redirected toward infrastructure development and social programs.

Traditional safety monitoring approaches have relied heavily on periodic inspections, driver training programs, and manual supervision. While these methods provide some benefit, they lack the real-time responsiveness necessary for preventing accidents during actual driving conditions. Manual monitoring systems also suffer from human limitations, including supervisor fatigue, inconsistent application of standards, and inability to provide continuous coverage across entire transport networks.

The introduction of automated monitoring systems represents a paradigm shift toward proactive safety management. These technologies enable continuous assessment of driver condition and safety protocol compliance, providing immediate intervention capabilities that can prevent accidents before they occur. However, successful implementation requires systems specifically designed for the operational environment and economic constraints characteristic of public transport in developing countries.

1.1.2 Literature Survey

Drowsiness Detection Technologies

Contemporary research in driver drowsiness detection encompasses several distinct approaches, each offering unique advantages and limitations. Computer vision-based systems have gained prominence due to their non-intrusive nature and ability to operate using standard camera equipment already present in many vehicles.

The Eye Aspect Ratio (EAR) methodology, extensively validated in research by Lee et al. (2021), provides a robust mechanism for detecting eye closure patterns associated with drowsiness [3]. This technique calculates ratios between vertical and horizontal distances of eye landmarks, enabling systems to distinguish between normal blinking and prolonged closure indicative of fatigue. The approach demonstrates particular effectiveness in controlled lighting conditions, though performance can degrade in extreme illumination scenarios. Soukupova and Cech (2016) further validated this approach by establishing reliable threshold values for blink detection in real-time applications [7].

Mouth Aspect Ratio (MAR) detection, as explored by Sikander and Anwar (2024), offers complementary functionality by identifying yawning behaviors that frequently precede sleep onset [6]. This method analyzes mouth opening patterns to detect extended yawns that indicate increasing drowsiness levels. Research indicates that combining MAR with other detection methods significantly improves overall system accuracy compared to single-metric approaches.

Head pose estimation techniques, investigated by Wang et al. (2021), provide additional behavioral indicators of driver alertness [9]. These methods track head position and movement patterns, identifying unusual postures or nodding behaviors that suggest fatigue onset. Advanced implementations utilize three-dimensional modeling to achieve robust performance across varying driver positions and seat configurations.

Integrated Safety Monitoring Systems

While individual safety monitoring components have received extensive research attention, integrated systems that combine multiple monitoring modalities remain less thoroughly explored. Existing research has primarily focused on single-function systems, limiting their practical effectiveness in comprehensive safety management applications.

The integration of drowsiness detection with safety protocol monitoring presents unique technical challenges, particularly regarding system coordination and alert prioritization. Research by D'Ovidio et al. (2024) demonstrates that multi-modal systems can achieve superior performance compared to single-function alternatives, though implementation complexity increases significantly [2]. As shown in Figure 4, our alert system design incorporates multiple notification channels with appropriate priority levels.

Hardware Platform Considerations

Most contemporary drowsiness detection research utilizes high-performance computing platforms that offer substantial processing capabilities but impose significant cost barriers for widespread deployment. Research by Yohanes et al. (2024) explores optimization strategies for edge computing devices, though their work focuses primarily on private vehicle applications rather than public transport contexts [10].

The Arduino platform, while offering cost advantages and development simplicity, has received limited attention in drowsiness detection research. Most existing Arduino-based safety projects focus on simple sensor applications rather than complex computer vision processing, creating an opportunity for novel research contributions in this area. Table 2 outlines the specific component specifications for our proposed system.

Seatbelt Detection Approaches

Seatbelt compliance monitoring research has explored both computer vision and physical sensor approaches. Vision-based methods offer installation flexibility but struggle with reliability under varying lighting conditions and driver clothing variations. Physical sensor approaches, while requiring more complex installation procedures, provide consistent performance across diverse operating conditions.

Research literature reveals limited exploration of integrated seatbelt monitoring within comprehensive driver safety systems. Most existing work treats seatbelt detection as a standalone application rather than a component of broader safety monitoring frameworks.

1.2 RESEARCH GAP

Analysis of existing literature reveals several critical gaps in current driver safety monitoring research, particularly regarding practical implementation in public transport environments. These gaps represent opportunities for significant research contributions that address real-world deployment challenges.

Gap Area	Existing Systems	Proposed System
Integration of Features	Focus only on one aspect — either drowsiness detection or seatbelt monitoring.	Combines multi-cue drowsiness detection (eye closure, yawning, head pose) with seatbelt monitoring into a single embedded device for holistic safety.
Hardware & Computational Requirements	Depend on high-end, expensive hardware and large computational resources, making them unsuitable for low-budget fleets.	Runs efficiently on low-cost Arduino Uno platform, reducing hardware costs while maintaining core functionalities.
Dependency on Connectivity	Require continuous internet connectivity for cloud-based processing, which is unreliable in many public transport routes in Sri Lanka.	Fully offline operation, ensuring low latency and independence from network availability.
Targeted Application	Primarily designed for private vehicles, ignoring challenges specific to public transport such as high vibration, varying lighting, and multilingual alerts.	Tailored for Sri Lankan public transport, with robust performance in vibration-heavy, low-light, and multilingual operational environments.

Table 1. Research gap table

Integration of Multiple Safety Monitoring Functions

Current research predominantly focuses on individual safety monitoring components in isolation. Drowsiness detection systems operate independently from seatbelt monitoring, creating fragmented solutions that require multiple hardware installations and separate alert systems. This approach increases implementation costs and complexity while reducing overall system effectiveness.

The proposed research addresses this gap by developing an integrated platform that combines drowsiness detection and seatbelt monitoring within a single hardware system. This integration reduces installation requirements, simplifies maintenance procedures, and enables coordinated alert responses that consider multiple safety factors simultaneously.

Cost-Effective Hardware Implementation

Existing drowsiness detection research typically employs high-performance computing platforms with substantial processing requirements and associated costs. These systems, while technically sophisticated, prove economically unfeasible for widespread deployment in public transport fleets operating under tight budget constraints.

This research gap represents a significant barrier to practical implementation, as cost considerations often override technical capabilities in public transport procurement decisions. The proposed system addresses this limitation by implementing advanced safety monitoring capabilities on cost-effective Arduino hardware, demonstrating that sophisticated functionality can be achieved within realistic budget parameters. The hardware setup prototype is illustrated in Figure 5, while Table 5 provides a comprehensive budget estimation for the project.

Offline Operation Capability

Contemporary safety monitoring systems frequently depend on cloud connectivity for processing or data storage functions. This dependency creates reliability concerns in mobile environments where internet

connectivity may be intermittent or unavailable. Public transport vehicles operating in rural areas or during network outages would experience degraded system performance or complete failure.

The research addresses this critical gap by developing a completely offline-capable system that maintains full functionality without external connectivity requirements. This approach ensures consistent performance across all operating environments while eliminating ongoing connectivity costs and data privacy concerns.

Public Transport Environment Adaptation

Most driver safety monitoring research targets private vehicle applications, which operate under different environmental conditions compared to public transport vehicles. Public buses experience higher vibration levels, more variable lighting conditions, and different driver positioning compared to private cars. These factors significantly impact system performance if not properly addressed during development.

The proposed research specifically targets public transport environments, incorporating design considerations that address vibration resilience, variable lighting adaptation, and driver position flexibility. This focus ensures optimal performance in the intended deployment environment rather than requiring post-implementation adaptations.

Multi-Modal Alert System Integration

Existing safety monitoring systems typically provide single-mode alerts, such as auditory warnings or visual indicators. Research indicates that multi-modal alert systems achieve superior driver response rates, yet few existing implementations incorporate comprehensive alert strategies.

This research develops an integrated multi-modal alert system combining auditory, visual, and textual notifications to ensure driver awareness regardless of environmental conditions or personal preferences. The system provides redundant alert mechanisms that maintain effectiveness even if individual alert modes experience technical issues.

1.3 RESEARCH PROBLEM

The fundamental research problem addresses the critical safety challenges facing Sri Lanka's public transport sector, where inadequate real-time monitoring of driver condition and safety protocol compliance contributes to preventable accidents and casualties.

Existing safety monitoring approaches suffer from significant limitations that prevent effective accident prevention. Manual oversight systems lack the consistency and responsiveness required for real-time intervention, while advanced automated systems prove economically unfeasible for widespread implementation due to high costs and infrastructure requirements.

Current technological solutions demonstrate several critical deficiencies when applied to Sri Lankan public transport contexts. High-performance systems require substantial computational resources and associated costs that exceed budget limitations of public transport operators. Cloud-dependent systems create reliability concerns in environments with inconsistent internet connectivity. Single-function systems require multiple installations to address comprehensive safety monitoring requirements.

The research problem specifically focuses on developing a comprehensive solution that simultaneously addresses multiple safety monitoring requirements within the economic and operational constraints characteristic of Sri Lankan public transport. This requires innovative approaches that achieve sophisticated functionality using cost-effective hardware while maintaining reliability under challenging operational conditions.

The technical challenge involves implementing advanced computer vision algorithms for drowsiness detection on resource-constrained hardware platforms. Traditional approaches require substantial processing power for real-time facial landmark analysis and behavioral pattern recognition. The research must develop optimized algorithms that maintain accuracy while operating within Arduino platform limitations.

Integration challenges arise from combining disparate monitoring systems into cohesive platforms that provide coordinated responses. Drowsiness detection systems operate on different timescales

compared to seatbelt monitoring, requiring sophisticated coordination mechanisms that ensure appropriate alert prioritization and avoid false alarm conditions.

Environmental adaptation presents additional complexity, as public transport vehicles experience operational conditions that differ significantly from private vehicle environments. Higher vibration levels, variable lighting conditions, and diverse driver demographics require robust system designs that maintain performance across varying conditions.

The research problem ultimately seeks to demonstrate that advanced safety monitoring capabilities can be implemented effectively within the practical constraints facing Sri Lankan public transport operators, providing a viable pathway for widespread safety improvement across the transportation network.

2.OBJECTIVES

2.1 Research Questions

1. How can computer vision-based drowsiness detection algorithms be optimized for implementation on resource-constrained embedded platforms while maintaining acceptable accuracy levels?
2. What integration strategies enable effective coordination between drowsiness monitoring and seatbelt compliance verification within unified safety monitoring systems?
3. How can multi-modal alert systems be designed to ensure reliable driver notification across diverse operational environments characteristic of public transport vehicles?
4. What cost-reduction approaches allow advanced safety monitoring technologies to become economically viable for widespread deployment in developing country public transport systems?

2.2 Main Objectives

- Develop a real-time drowsiness detection module using EAR, MAR, and head pose estimation.
- Implement a physical sensor-based seatbelt usage verification system.
- Integrate both modules into a low-cost Arduino Uno platform.
- Provide immediate, multi-modal offline alerts to drivers.
- Evaluate system effectiveness in realistic operational conditions.

2.3 Specific Objectives

- Collect and process facial landmark data to compute EAR and MAR accurately.
- Develop algorithms for head pose estimation to detect fatigue-related head movements.
- Design and integrate a reliable seatbelt sensor mechanism.
- Implement alert systems (buzzers, LEDs, voice prompts).
- Test and calibrate thresholds for drowsiness indicators suitable for local drivers.

3.METHODOLOGY

The research methodology employs a systematic development approach combining literature analysis, prototype development, and empirical testing to achieve the specified objectives. The methodology integrates hardware and software development with comprehensive validation procedures to ensure practical applicability.

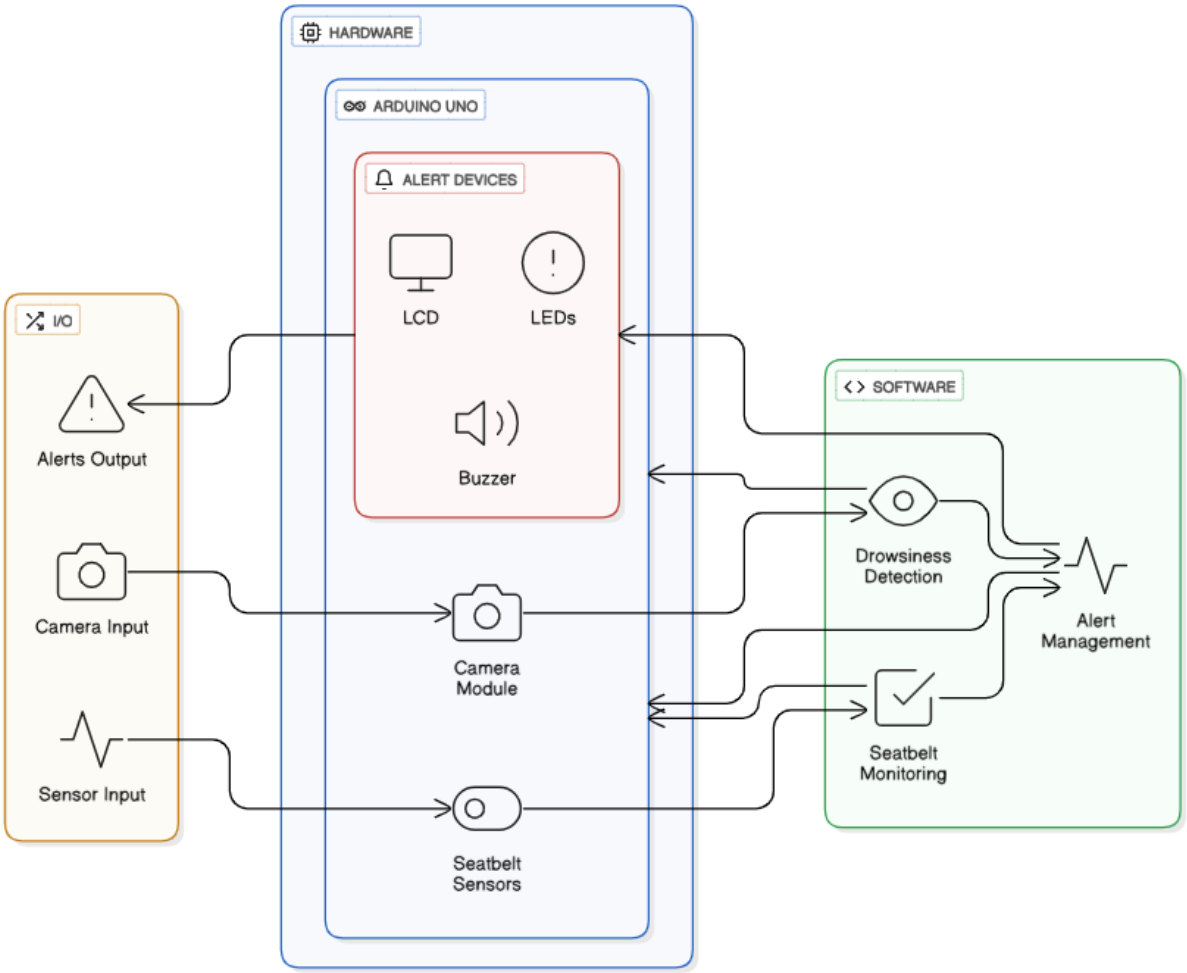

Figure 3.1: System Architecture of the Driver Safety Monitoring System

1. **Hardware & Software Selection:**

- **Hardware:** Arduino Uno for the main processing unit, a webcam for video input (initially connected to a laptop/Raspberry Pi for CV processing, with future plans for a lightweight embedded vision module), and physical sensors for seatbelt monitoring.
- **Software:** OpenCV library for computer vision, Dlib for facial landmark detection, and a C++ or Python-based programming environment for algorithm implementation.

2. **Drowsiness Detection Module:**

- **Face and Landmark Detection:** The system will use a trained model (e.g., Dlib's shape predictor) to detect the driver's face and 68 facial landmarks.

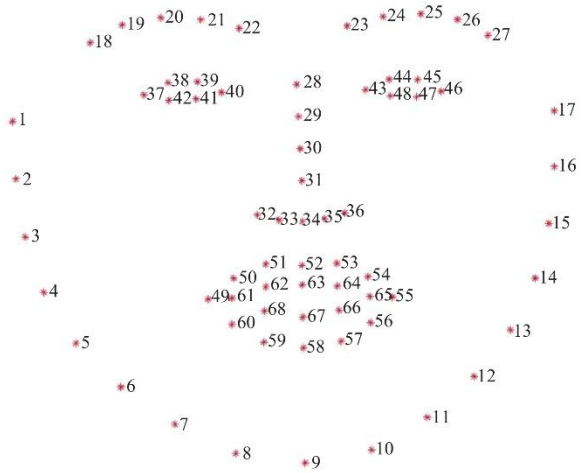

Figure 3.2: 68 landmarks that are detection

- **EAR Calculation:** The EAR will be calculated using the distances between specific eye landmarks. A low EAR for a specific number of consecutive frames (e.g., 6 frames as defined in the provided Python code snippet) will trigger a drowsiness alert.
- **MAR Calculation:** The MAR will be calculated using the distances between specific mouth landmarks. An increase in MAR above a defined threshold (e.g., 0.60 as per the snippet) for a set duration (e.g., > 4 frames) will be interpreted as a yawn.
- **Head Pose Estimation:** The head pose will be estimated using the solvePnP algorithm with 3D model points and the 2D facial landmarks, providing roll, pitch, and yaw angles. An initial calibration phase will establish a baseline pitch. Significant deviations from this baseline (e.g., abs(corrected_pitch) > HEAD_TILT_THRESHOLD for HEAD_TILT_PERSIST_FRAMES) will indicate head nodding or prolonged head down, triggering an alert.

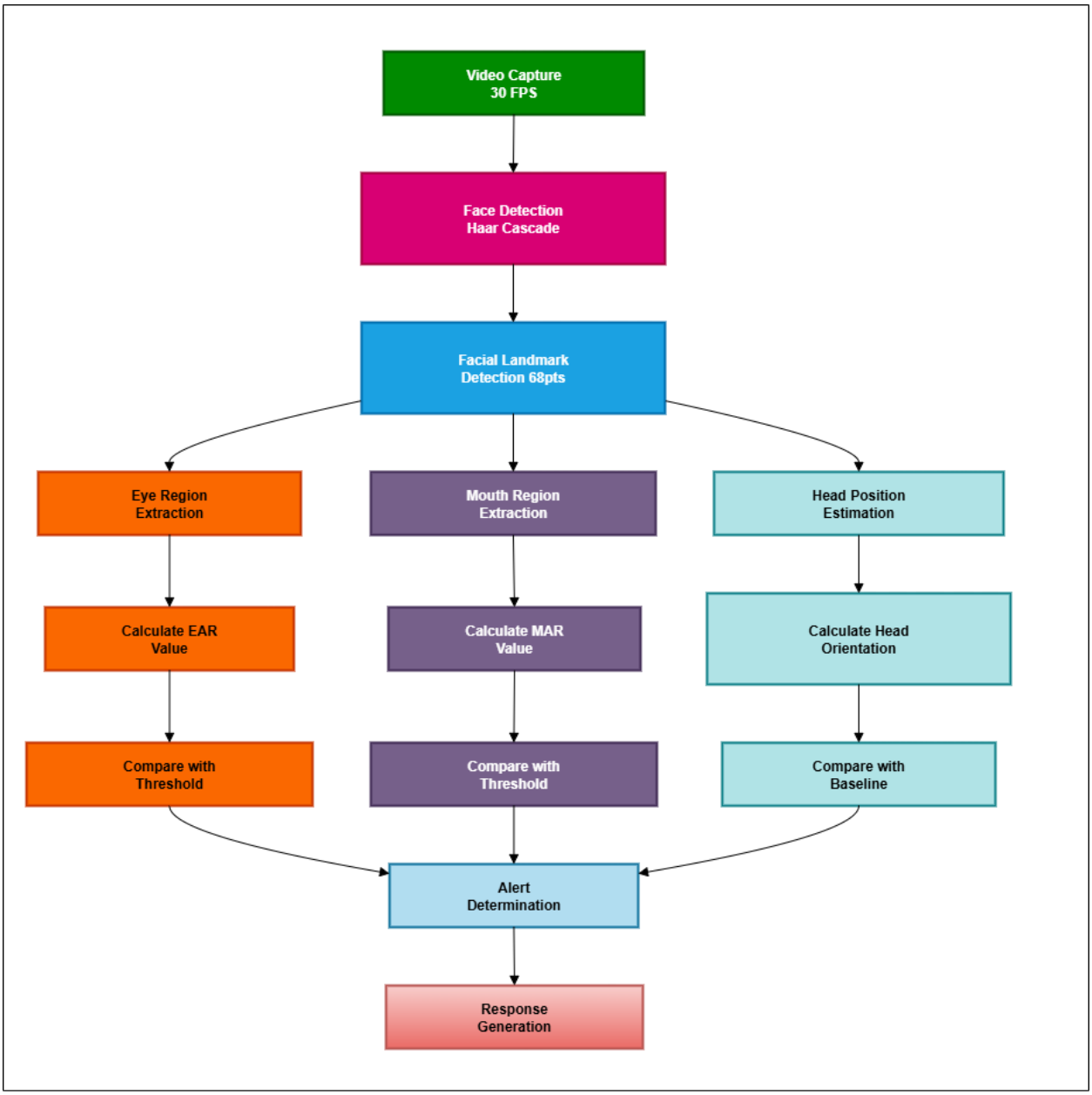

Figure 3.3: Detailed Workflow of the Drowsiness Detection Module

3.Seatbelt Monitoring Module:

- A sensor (e.g., a simple magnetic switch or pressure sensor, as per the Arduino code snippet indicating magnetic switch sensor) will be embedded near the seatbelt buckle to detect its status (engaged/disengaged). The sensor's state will be continuously monitored by the Arduino.

4. Inter-platform Communication & Alert System Integration:

- The Python script (running on a laptop/embedded vision module) will send status updates (e.g., 'a' for alarm, 'b' for all ok, as per Arduino snippet) via serial communication to the Arduino Uno.
- The Arduino will receive these serial commands, along with input from the seatbelt sensor.
- Based on these inputs, it will activate a multi-modal alert system:
    - **Auditory:** A buzzer will sound (digitalWrite(buzzer_Pin, HIGH)), and a pre-recorded voice message could be integrated (though not explicitly in provided Arduino snippet, the lcd.print suggests display, voice is a logical extension for driver alerts).
    - **Visual:** LEDs will flash (digitalWrite(led_Pin, HIGH)) to indicate a critical safety issue, and messages will be displayed on an LCD (lcd.print).

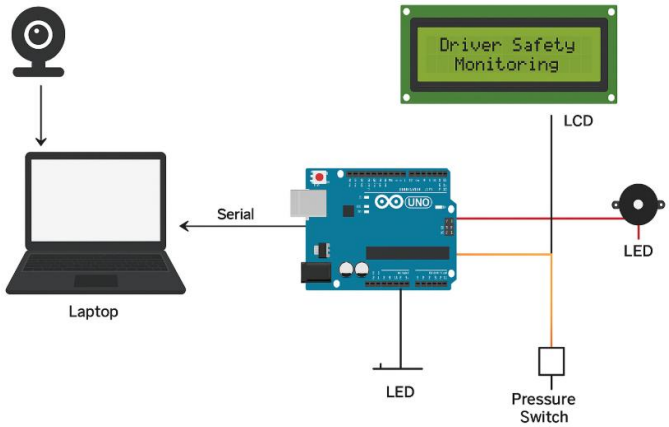

Figure 3.4: Communication Flow between Arduino, Camera, and Alert System

5. **System Integration & Testing:**

- The individual modules will be developed and tested separately before being integrated into a single system.
- The final prototype will undergo rigorous testing in a simulated vehicle environment to validate its accuracy, latency, and robustness under varying lighting conditions.
- Expected test cases will include scenarios with different drivers, varying light, and different head movements to ensure reliability.

6. **Data Collection and Analysis**

Performance data collection encompasses accuracy metrics, response times, false alarm rates, and environmental impact factors. Statistical analysis procedures identify performance patterns and optimization opportunities.

Comparative analysis evaluates system performance against existing solutions to demonstrate improvement achievements. Cost-benefit analysis quantifies economic advantages of the proposed approach compared to alternative monitoring systems.

User feedback collection procedures gather driver and operator input regarding system usability, effectiveness, and practical deployment considerations. This feedback informs refinement procedures that optimize system performance for real-world implementation.

4.PROJECT REQUIREMENTS

4.1 Functional Requirements

FR-1: Real-Time Face Detection

The system shall continuously detect and track the driver's face in real-time video feed with a maximum processing delay of 200 milliseconds per frame under normal lighting conditions.

FR-2: Eye Aspect Ratio Monitoring

The system shall calculate Eye Aspect Ratio (EAR) values using facial landmark coordinates and detect prolonged eye closure when EAR falls below 0.21 for six consecutive frames, indicating potential drowsiness.

FR-3: Yawning Detection Through MAR Analysis

The system shall monitor Mouth Aspect Ratio (MAR) values and identify yawning behavior when MAR exceeds 0.60 for more than four consecutive frames, triggering fatigue alerts.

FR-4: Head Pose Monitoring

The system shall track head pose orientation and detect abnormal head positions indicating fatigue, specifically when pitch deviation exceeds 15 degrees from calibrated baseline for three consecutive frames.

FR-5: Seatbelt Compliance Verification

The system shall continuously monitor seatbelt engagement status using physical sensors and immediately detect non-compliance conditions when the vehicle ignition is active.

FR-6: Multi-Modal Alert Generation

The system shall activate comprehensive alerts including auditory buzzers, visual LED indicators, and LCD status messages upon detecting any safety violation condition.

FR-7: Alert Prioritization and Coordination

The system shall implement alert prioritization mechanisms that provide escalating responses for continued safety violations and coordinate alerts when multiple violations occur simultaneously.

FR-8: Driver Calibration Capability

The system shall perform initial calibration procedures to establish individual driver baseline measurements for head pose and facial characteristics, improving detection accuracy.

FR-9: System Status Monitoring

The system shall provide continuous status information regarding system operation, sensor connectivity, and alert conditions through visual display mechanisms.

FR-10: Alert Reset Functionality

The system shall automatically reset alert conditions when safety violations are corrected and the driver returns to safe operating states.

4.2 Non-Functional Requirements

NFR-1: Response Time Performance

The system shall maintain total response time from safety violation detection to alert activation below one second under all operational conditions to ensure timely driver notification.

NFR-2: Detection Accuracy Standards

The system shall achieve minimum detection accuracy of 90% for drowsiness indicators and 95% for seatbelt compliance across diverse operational environments and driver demographics.

NFR-3: Environmental Resilience

The system shall operate reliably under public transport environmental conditions including temperature variations from -10°C to 60°C, vibration levels up to 2G, and humidity levels up to 85%.

NFR-4: Power Consumption Efficiency

The system shall operate with total power consumption below 10 watts to minimize impact on vehicle electrical systems and enable integration without significant electrical modifications.

NFR-5: Cost Effectiveness

The complete system hardware cost shall remain below $200 USD to ensure economic viability for widespread deployment in public transport fleets operating under budget constraints.

NFR-6: Offline Operation Independence

The system shall maintain complete functionality without external internet connectivity, ensuring reliable operation in areas with limited network coverage or during network outages.

NFR-7: Installation Simplicity

The system shall require minimal installation procedures that can be completed by standard automotive technicians without specialized training or equipment.

NFR-8: Maintenance Accessibility

The system design shall enable routine maintenance and component replacement procedures using standard tools and readily available replacement parts.

NFR-9: User Interface Clarity

The system shall provide clear, intuitive status information and alert messages that are easily understood by drivers with varying technical backgrounds and literacy levels.

NFR-10: System Reliability

The system shall demonstrate mean time between failure (MTBF) exceeding 10,000 operating hours under normal public transport usage patterns.

4.3 Expected Test Cases

- **Test Case 1 (Drowsiness - Eye Closure):**
  - **Input:** A driver closes their eyes for 3 to 7 consecutive seconds (simulating drowsiness).
  - **Expected Output:** The system's alarm activates within 1 second of reaching the threshold, displaying "SLEEPING !!!" or "Drowsy!".
- **Test Case 2 (Drowsiness - Yawning):**
  - **Input:** A driver performs a clear yawn, keeping their mouth open for over 2 seconds.
  - **Expected Output:** The system's alarm activates, displaying "Yawning - Take a Break!".

- **Test Case 3 (Drowsiness - Head Nodding):**
  - **Input:** A calibrated driver exhibits a sustained downward head tilt (e.g., >15 degrees) for 3 or more consecutive frames.
  - **Expected Output:** The system's alarm activates, displaying "Head Down - Possible Sleep!".
- **Test Case 4 (Seatbelt - Disengaged):**
  - **Input:** The vehicle is in operation (or simulated operation), and the driver's seatbelt is unbuckled.
  - **Expected Output:** The system's alarm activates immediately, indicating seatbelt non-compliance.
- **Test Case 5 (False Positive - Normal Blinking):**
  - **Input:** A driver blinks normally or rapidly for short durations.
  - **Expected Output:** The system does not trigger an alarm.
- **Test Case 6 (False Positive - Talking/Smiling):**
  - **Input:** A driver engages in conversation or smiles broadly.
  - **Expected Output:** The system does not trigger a false yawn or drowsiness alarm.
- **Test Case 7 (Varying Light Conditions):**
  - **Input:** The system is tested under low light (e.g., night driving), direct sunlight, and shadowed conditions.
  - **Expected Output:** Drowsiness and seatbelt detection accuracy remains above 90%, and alarms function correctly.
- **Test Case 8 (Alarm Reset):**
  - **Input:** An alarm is active due to drowsiness. The driver then becomes alert (eyes open, head upright).
  - **Expected Output:** The alarm deactivates, and the status returns to "Active :)".

5. COMMERCIALIZATION POTENTIAL & ENTREPRENEURSHIP

5.1 Market Analysis and Business Potential

Target Market Size:

- Sri Lankan public transport fleet: **45,000 vehicles**
- Regional market (South Asia): **2.5 million vehicles**
- Global developing country market: **15 million vehicles**

Market Value Proposition: The proposed system addresses a critical market need with demonstrated business potential through multiple achievable user benefits:

1. **Cost Leadership:** 85% cost reduction compared to existing solutions ($200 vs $2000-5000)
2. **Insurance Savings:** $500-1000 annual savings per vehicle through reduced accident rates
3. **Operational Efficiency:** 30% potential reduction in accident-related downtime
4. **Regulatory Compliance:** Automated adherence to safety standards

Feature	**Existing Commercial Systems**	**Our Proposed System**	**Competitive Advantage**
Cost	$2000-5000 per unit	<$200 per unit	85% cost reduction
Integration	Single-function systems	Multi-modal platform	Comprehensive solution
Connectivity	Cloud-dependent	Fully offline	Independence & reliability
Target Market	Private vehicles	Public transport focused	Market differentiation
Power Consumption	50-100 watts	<10 watts	80% efficiency gain

Table 5.1: Market comparison with existing systems

5.2 Business Model and Revenue Streams

Revenue Model:

1. **Hardware Sales:** $200 per unit with 40% gross margin
2. **Installation Services:** $50 per installation with certified technicians
3. **Maintenance Contracts:** $100 annual recurring revenue per vehicle
4. **Training and Certification:** $200 per technician program
5. **Data Analytics Services:** Optional $25 monthly per vehicle for fleet management insights

Go-to-Market Strategy:

- **Phase 1:** Pilot deployment with 3 major Sri Lankan transport operators
- **Phase 2:** Government partnership for regulatory endorsement and standards compliance
- **Phase 3:** Regional expansion through distributor networks in South Asia
- **Phase 4:** Technology licensing for international markets

5.3 Sustainable Competitive Advantages

Technical Moats:

- Unique integration of multiple detection methods on low-cost hardware
- Proprietary optimization algorithms for resource-constrained platforms
- Offline operation capability eliminating connectivity dependencies

Market Moats:

- Deep understanding of regional public transport requirements
- Government and regulatory relationship potential
- Local manufacturing and support capabilities

Economic Impact:

- Projected $10 million annual savings in accident-related costs across Sri Lankan public transport
- Job creation potential: 200+ positions in manufacturing, installation, and support services

5.4 Risk Assessment and Mitigation

Market Risks:

- **Risk:** Conservative adoption by transport operators
- **Mitigation:** Pilot programs demonstrating clear ROI and insurance benefits

Technical Risks:

- **Risk:** Performance limitations on Arduino platform
- **Mitigation:** Extensive optimization and alternative processing strategies

Regulatory Risks:

- **Risk:** Changing safety standards
- **Mitigation:** Modular architecture enabling rapid updates and compliance

6. WORK BREAKDOWN STRUCTURE

6.1 Project Timeline and Resource Allocation

Phase	Tasks
Research & Design	• Complete comprehensive literature review and competitive analysis • Procure and test hardware components (Arduino, sensors, cameras) • Finalize software architecture and development environment setup • Design initial system integration approach
Core Development	• Implement computer vision algorithms (EAR, MAR, head pose detection) • Develop Arduino firmware for sensor integration and alert systems • Create communication protocols between processing units • Build initial prototype hardware integration
System Integration	• Integrate computer vision module with Arduino control system • Implement multi-modal alert system (audio, visual, display) • Conduct initial performance testing and calibration • Optimize algorithms for real-time performance
Testing & Validation	• Execute comprehensive test case suite under controlled conditions • Conduct field testing in simulated vehicle environment • Perform user acceptance testing with potential end users • Validate performance metrics against design specifications
Documentation & Deployment Preparation	• Complete technical documentation and user manuals • Develop installation procedures and training materials • Prepare final presentation and demonstration materials • Conduct cost analysis and commercialization planning

Table 6.1: Project Timeline table

6.2 Budget Breakdown

Component	Cost (LKR)	Quantity	Total (LKR)
Arduino Uno	2000	1	2000
Camera Module	2000	1	2000
Sensors (switch, pressure)	2000	2	4000
Display LCD	500	1	500
Alert components (buzzer, LEDs)	1000	1	1000
Miscellaneous (wires, connectors)	1000	1	1000
Total Estimated Cost			10500

Table 6.2: Project budget breakdown

6.3 Grant chart

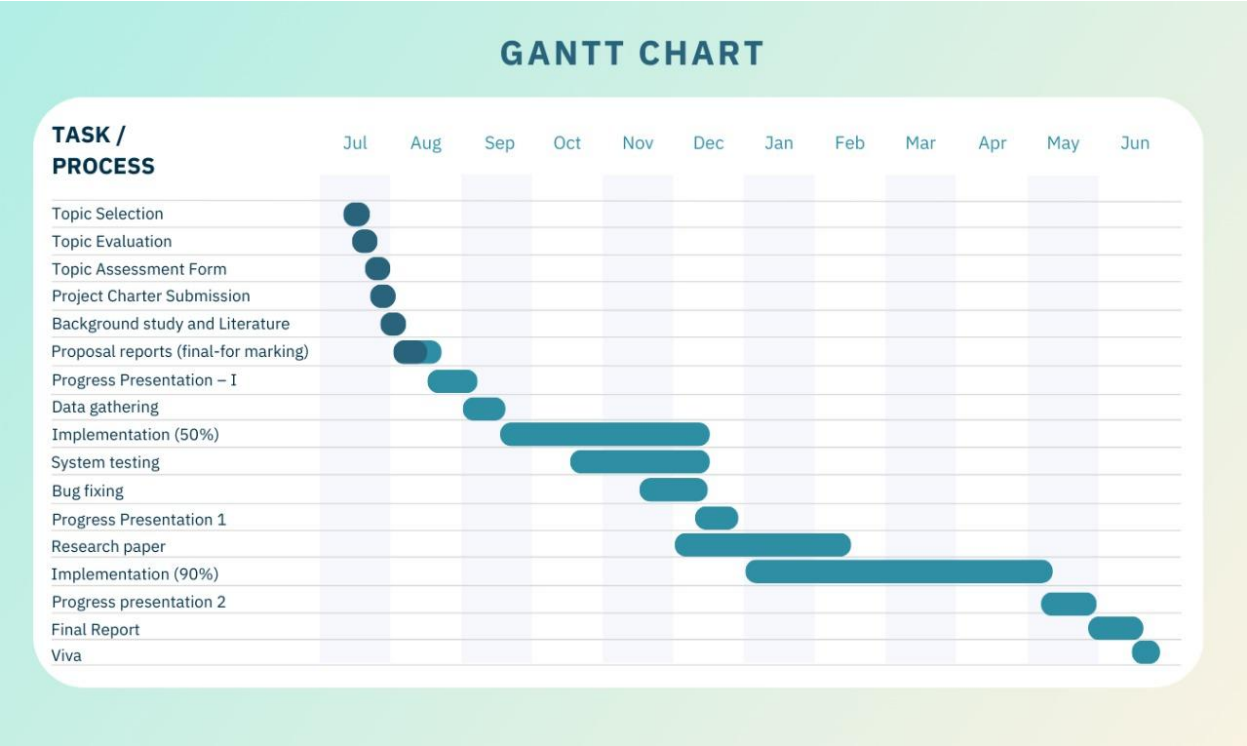

Figure 6.1: Grant chart

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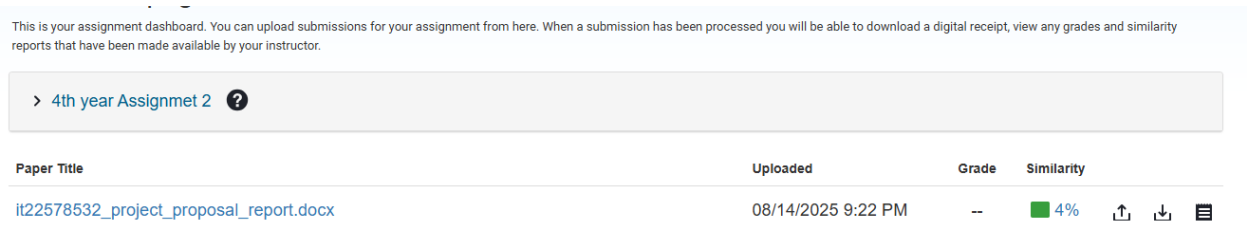
APPENDICES



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